

Question	Answer	Marks
2(a)(i)	p = pressure (of gas), V = volume (of gas) and k = Boltzmann constant	B1
	N = number of molecules (in the gas)	B1
	T = thermodynamic temperature (of gas)	B1
2(a)(ii)	$(pV =) NkT = \frac{1}{3}Nm\langle c^2 \rangle$	M1
	$E_K = \frac{1}{2}m\langle c^2 \rangle = \frac{1}{2} \times 3kT = (3/2)kT$	A1
2(b)	(same temperature so) $(\frac{1}{2})m\langle c^2 \rangle$ must be same for both gases	C1
	ratio = $\sqrt{(m_O / m_H)} = \sqrt{(32 / 2)}$ = 4.0	A1

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